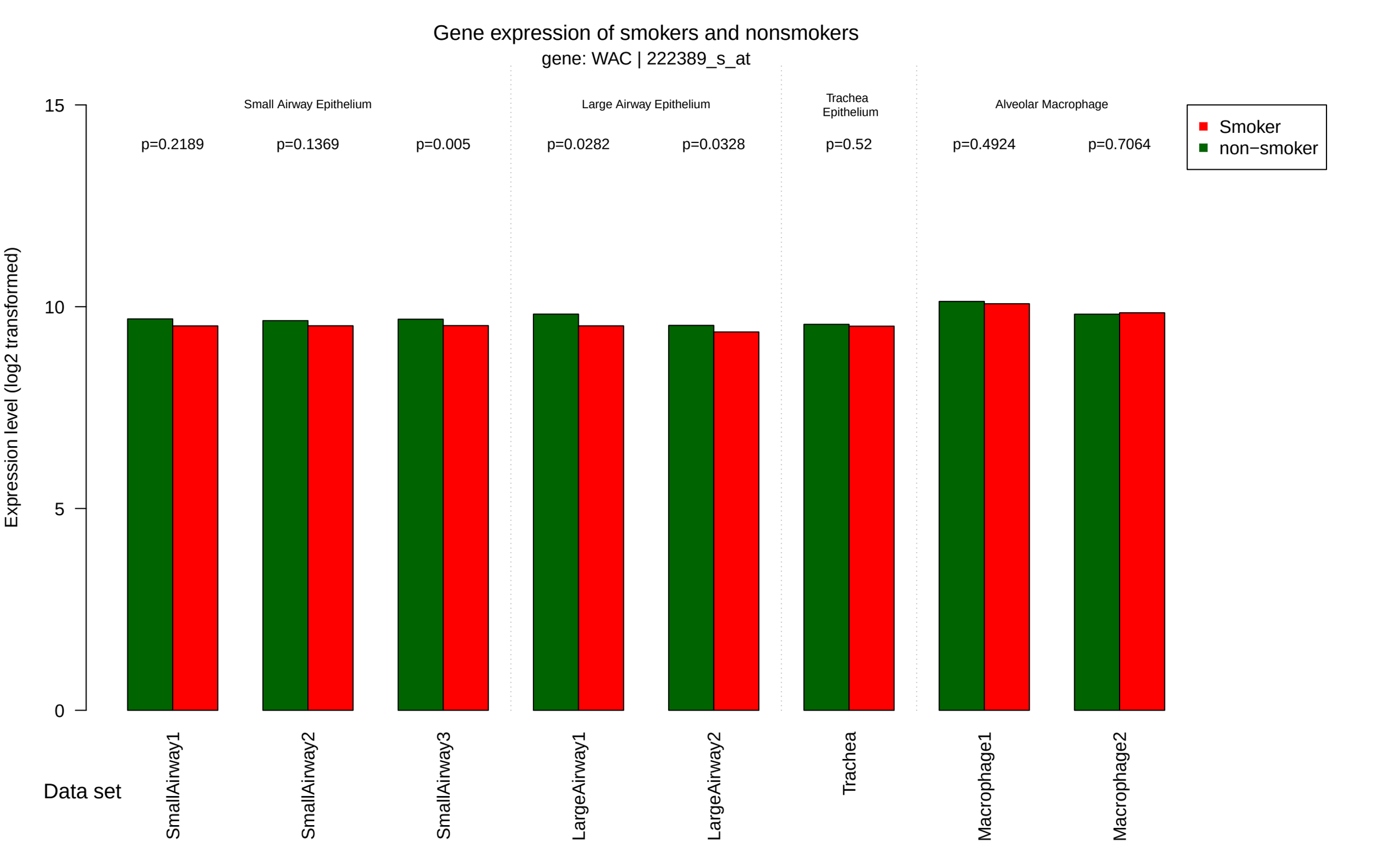
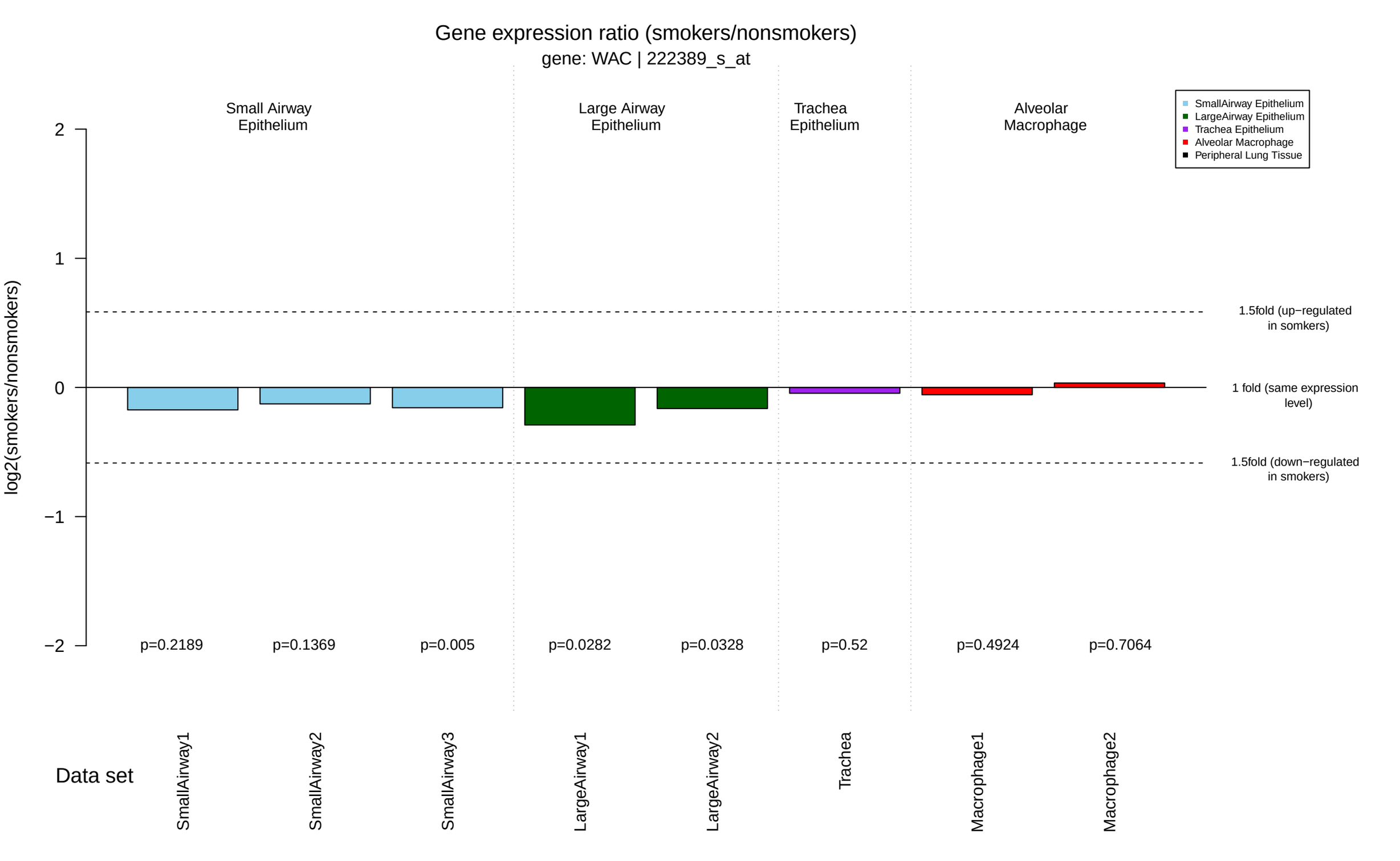
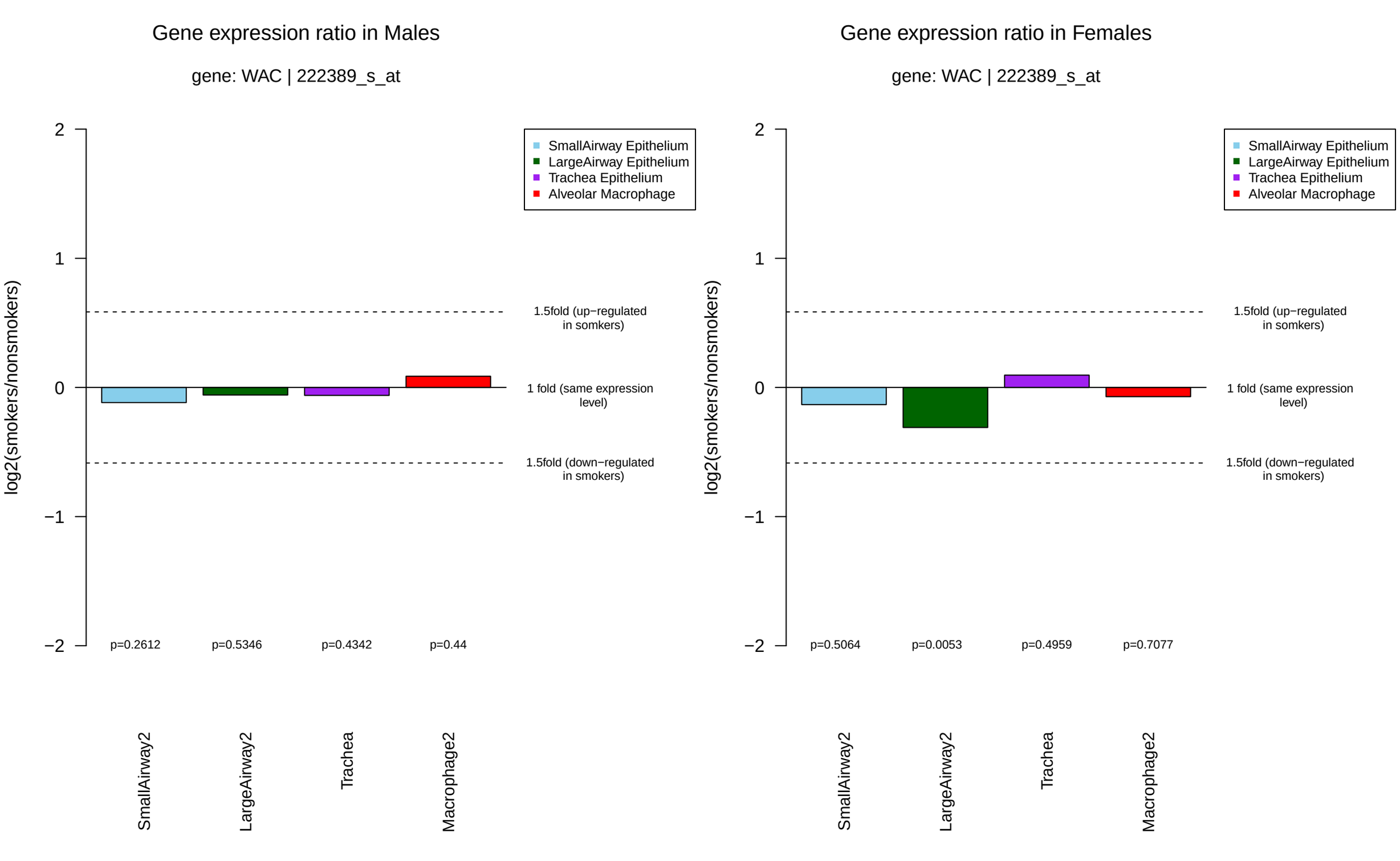
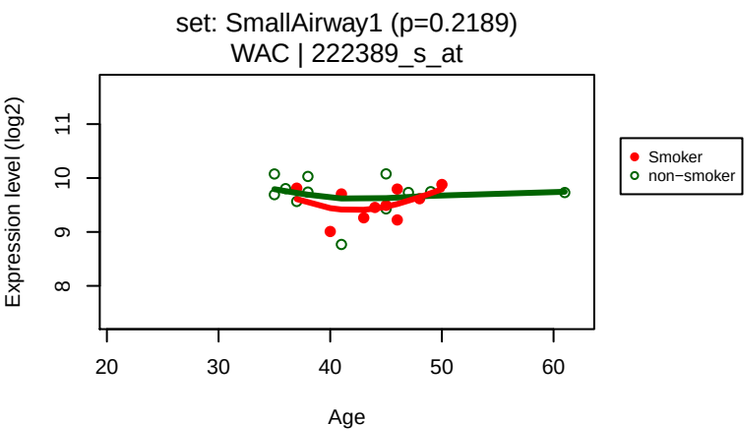
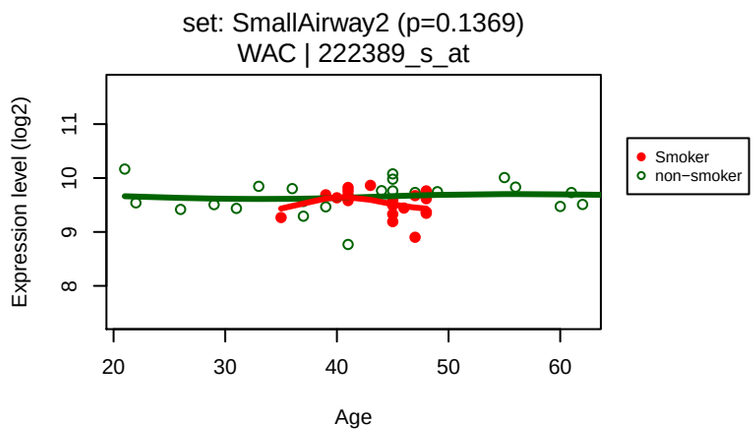
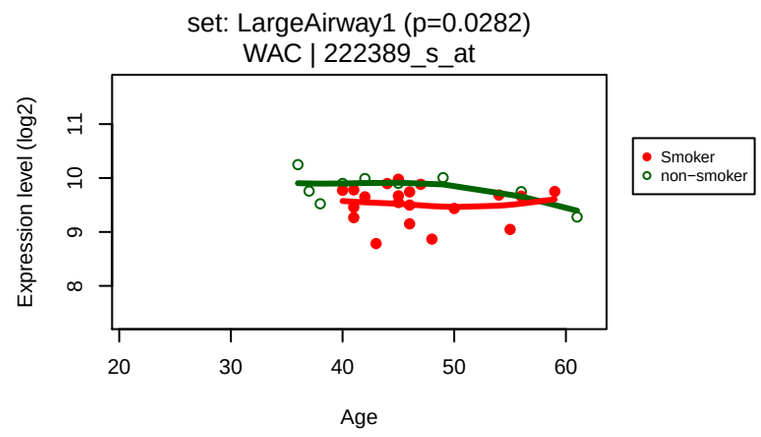

Gene expression of smokers and nonsmokers
gene: WAC | 222389_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2189
p=0.1369
p=0.005
p=0.0282
p=0.0328
p=0.52
p=0.4924
p=0.7064
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

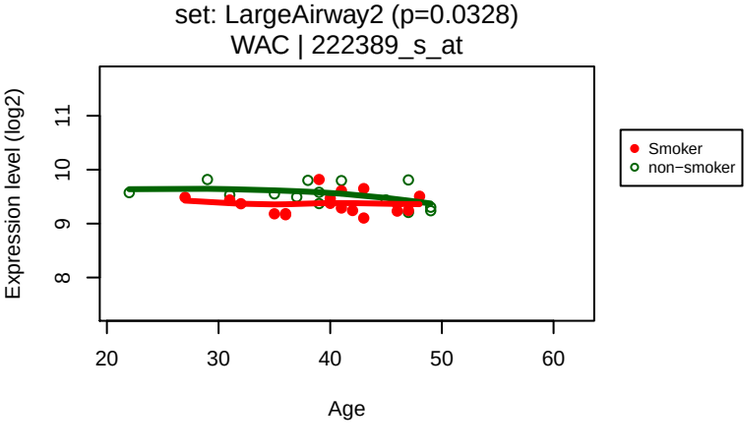

Gene expression ratio (smokers/nonsmokers)
gene: WAC | 222389_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2189
p=0.1369
p=0.005
p=0.0282
p=0.0328
p=0.52
p=0.4924
p=0.7064
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

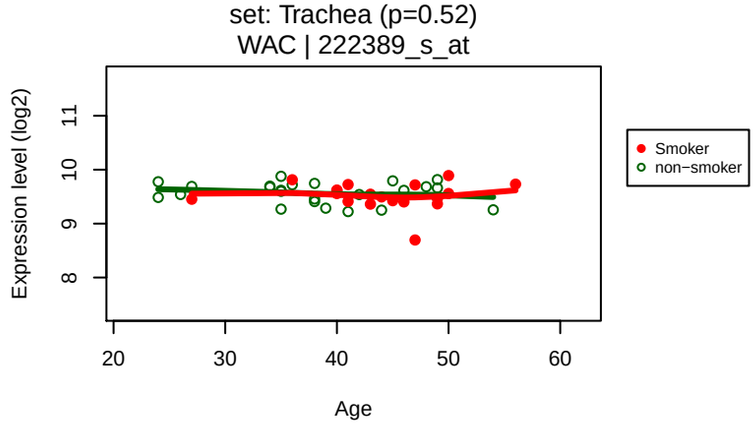

Gene expression ratio in Males
gene: WAC | 222389_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2612
p=0.5346
p=0.4342
p=0.44
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: WAC | 222389_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5064
p=0.0053
p=0.4959
p=0.7077
SmallAirway2
LargeAirway2
Trachea
Macrophage2

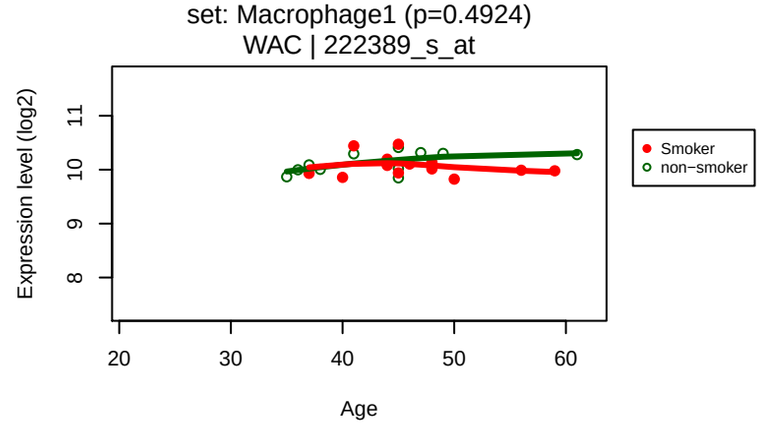

set: SmallAirway1 (p=0.2189)
WAC | 222389_s_at
Expression level (log2)
Age
Smoker
non−smoker

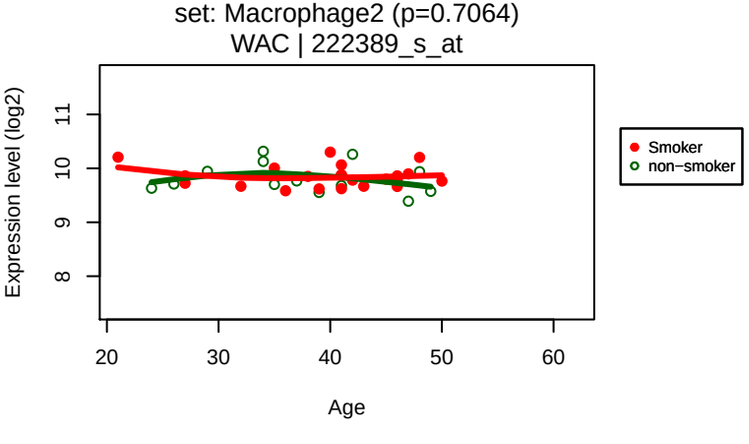

set: SmallAirway2 (p=0.1369)
WAC | 222389_s_at
Expression level (log2)
Age
Smoker
non−smoker

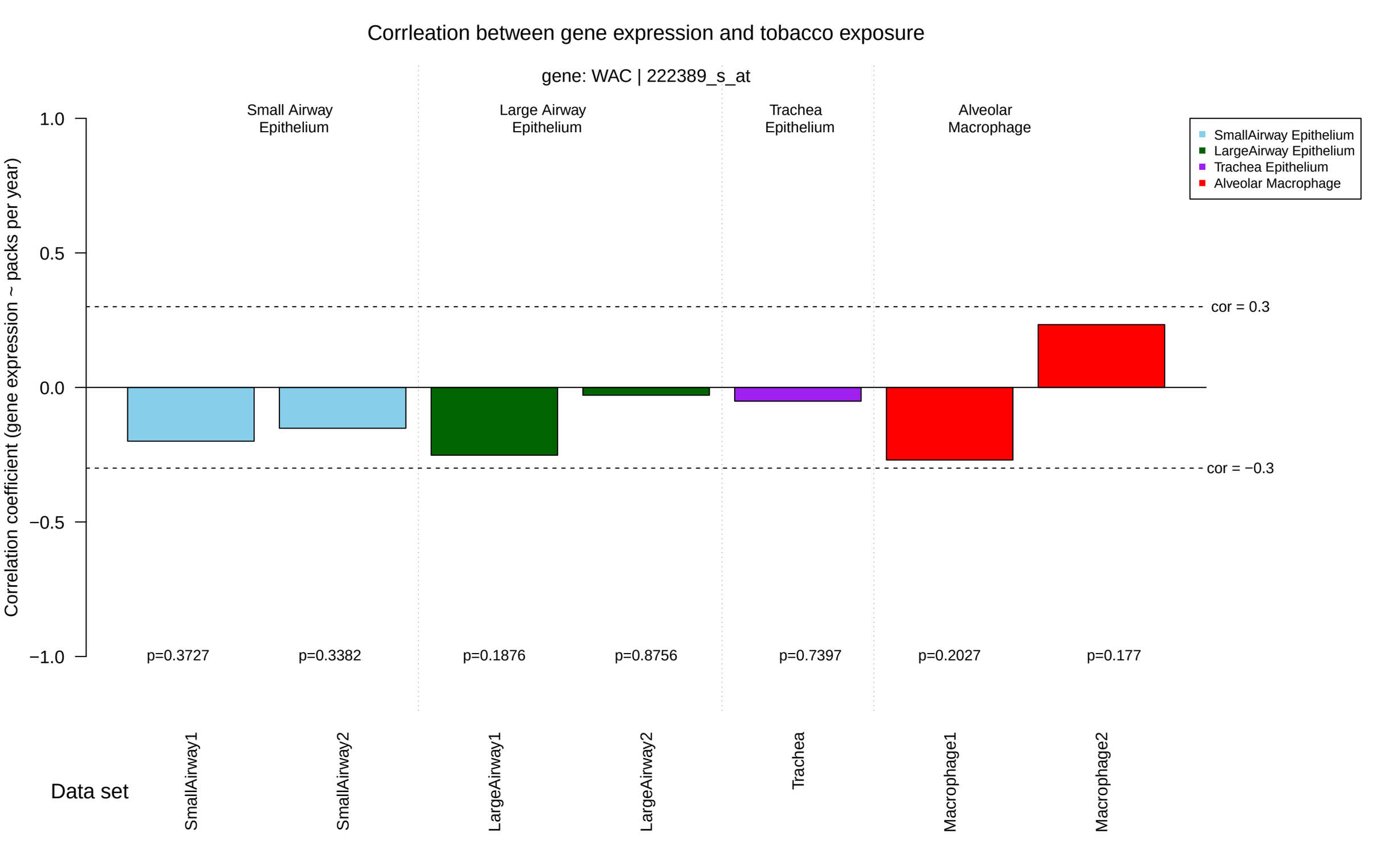

set: LargeAirway1 (p=0.0282)
WAC | 222389_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0328)
WAC | 222389_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.52)
WAC | 222389_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.4924)
WAC | 222389_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.7064)
WAC | 222389_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: WAC | 222389_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.3727
p=0.3382
p=0.1876
p=0.8756
p=0.7397
p=0.2027
p=0.177
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2